Tutorial Problems:Solutions

Problems with Detailed Solutions

Problem 1: Checking sigma-fields on a finite sample space

Let

$$\Omega = \{1, 2, 3, 4\}.$$

Determine which of the following collections is a sigma-field of subsets of Ω:

$$\begin{aligned}
\mathcal{F}_1 &= \big\{\varnothing, \{1, 2\}, \{3, 4\}\big\}, \\
\mathcal{F}_2 &= \big\{\varnothing, \Omega, \{1\}, \{2, 3, 4\}, \{1, 2\}, \{3, 4\}\big\}, \\
\mathcal{F}_3 &= \big\{\varnothing, \Omega, \{1\}, \{2\}, \{1, 2\}, \{3, 4\}, \\
&\qquad\qquad \{2, 3, 4\}, \{1, 3, 4\}\big\}.
\end{aligned}$$

Justify your answer by checking the sigma-field axioms.

Solution.

A collection $\mathcal{F} \subseteq 2^{\Omega}$ is a sigma-field if:

$$\Omega \in \mathcal{F},$$

$$A \in \mathcal{F} \quad \Longrightarrow \quad A^c \in \mathcal{F},$$

and

$$A_n \in \mathcal{F} \text{ for every } n \quad \Longrightarrow \quad \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Because Ω is finite, it is enough to check closure under complements and finite unions.

Checking $\mathcal{F}_1$.
We have

$$\Omega \notin \mathcal{F}_1.$$

Equivalently,

$$\varnothing \in \mathcal{F}_1, \qquad \varnothing^c = \Omega \notin \mathcal{F}_1.$$

Thus, $\mathcal{F}_1$ fails the complement condition. Therefore,

$$\boxed{\mathcal{F}_1 \text{ is not a sigma-field.}}$$

Checking $\mathcal{F}_2$.
The collection is closed under complements:

$$\varnothing^c = \Omega, \qquad \{1\}^c = \{2, 3, 4\}, \qquad \{1, 2\}^c = \{3, 4\}.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

However,

$$\{1\} \in \mathcal{F}_2 \quad \text{and} \quad \{3,4\} \in \mathcal{F}_2,$$

whereas

$$\{1\} \cup \{3,4\} = \{1,3,4\} \notin \mathcal{F}_2.$$

Therefore, $\mathcal{F}_2$ is not closed under finite unions. Hence,

$$\boxed{\mathcal{F}_2 \text{ is not a sigma-field.}}$$

Checking $\mathcal{F}_3$.

Consider the partition

$$\mathcal{P} = \big\{\{1\},\{2\},\{3,4\}\big\}.$$

The collection $\mathcal{F}_3$ contains all possible unions of the three blocks of this partition:

$$\begin{aligned}
&\varnothing,\\
&\{1\}, \quad \{2\}, \quad \{3,4\},\\
&\{1,2\}, \quad \{1,3,4\}, \quad \{2,3,4\},\\
&\Omega.
\end{aligned}$$

There are $2^3 = 8$ such unions, and all eight are present in $\mathcal{F}_3$.

The complement of a union of blocks is the union of the remaining blocks. Also, any union of sets formed from these blocks is again a union of blocks. Consequently, $\mathcal{F}_3$ is closed under complements and countable unions.

Therefore,

$$\boxed{\mathcal{F}_3 \text{ is a sigma-field.}}$$

Hence, the final answer is

$$\boxed{\text{Only } \mathcal{F}_3 \text{ is a sigma-field.}}$$

□

Problem 2: Intersection and union of two sigma-fields

Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two sigma-fields of subsets of the same sample space Ω.

(i) Prove that

$$\mathcal{F}_1 \cap \mathcal{F}_2$$

is always a sigma-field of subsets of Ω.

(ii) Is

$$\mathcal{F}_1 \cup \mathcal{F}_2$$

necessarily a sigma-field? Justify your answer using a counterexample.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution.

Part (i): Intersection of two sigma-fields.

We verify the three sigma-field conditions.

1. The sample space belongs to the intersection.

Since $\mathcal{F}_1$ and $\mathcal{F}_2$ are sigma-fields,

$$\Omega \in \mathcal{F}_1 \qquad \text{and} \qquad \Omega \in \mathcal{F}_2.$$

Therefore,

$$\Omega \in \mathcal{F}_1 \cap \mathcal{F}_2.$$

2. Closure under complements.

Let

$$A \in \mathcal{F}_1 \cap \mathcal{F}_2.$$

This means that

$$A \in \mathcal{F}_1 \qquad \text{and} \qquad A \in \mathcal{F}_2.$$

Because both collections are sigma-fields,

$$A^c \in \mathcal{F}_1 \qquad \text{and} \qquad A^c \in \mathcal{F}_2.$$

Hence,

$$A^c \in \mathcal{F}_1 \cap \mathcal{F}_2.$$

Thus, the intersection is closed under complements.

3. Closure under countable unions.

Let

$$A_n \in \mathcal{F}_1 \cap \mathcal{F}_2, \qquad n = 1, 2, \ldots.$$

Then every A_n belongs to both sigma-fields:

$$A_n \in \mathcal{F}_1 \qquad \text{and} \qquad A_n \in \mathcal{F}_2.$$

Since $\mathcal{F}_1$ and $\mathcal{F}_2$ are closed under countable unions,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_1$$

and

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_2.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Therefore,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_1 \cap \mathcal{F}_2.$$

All three sigma-field conditions are satisfied. Hence,

$$\boxed{\mathcal{F}_1 \cap \mathcal{F}_2 \text{ is always a sigma-field.}}$$

Part (ii): Union of two sigma-fields.

The union of two sigma-fields need not be a sigma-field.

Consider

$$\Omega = \{1, 2, 3\}$$

and define

$$\mathcal{F}_1 = \big\{\varnothing, \Omega, \{1\}, \{2, 3\}\big\}$$

and

$$\mathcal{F}_2 = \big\{\varnothing, \Omega, \{2\}, \{1, 3\}\big\}.$$

Both $\mathcal{F}_1$ and $\mathcal{F}_2$ are sigma-fields.

For $\mathcal{F}_1$, the complement pairs are

$$\varnothing^c = \Omega, \qquad \{1\}^c = \{2, 3\}.$$

For $\mathcal{F}_2$, the complement pairs are

$$\varnothing^c = \Omega, \qquad \{2\}^c = \{1, 3\}.$$

Both collections are also closed under countable unions.

Their union is

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \big\{\varnothing, \Omega, \{1\}, \{2\}, \{2, 3\}, \{1, 3\}\big\}.$$

Now,

$$\{1\} \in \mathcal{F}_1 \cup \mathcal{F}_2$$

and

$$\{2\} \in \mathcal{F}_1 \cup \mathcal{F}_2.$$

However,

$$\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{F}_1 \cup \mathcal{F}_2.$$

Thus, $\mathcal{F}_1 \cup \mathcal{F}_2$ is not closed under finite unions and therefore cannot be a sigma-field.

Hence,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$\boxed{\mathcal{F}_1 \cup \mathcal{F}_2 \text{ need not be a sigma-field.}}$

The final conclusion is

$\boxed{\begin{array}{l}\mathcal{F}_1 \cap \mathcal{F}_2 \text{ is always a sigma-field,} \\ \mathcal{F}_1 \cup \mathcal{F}_2 \text{ need not be a sigma-field.}\end{array}}$

□

Probability Measures

Problem 3: Checking probability measures on a countable sample space

Let

$$\Omega = \{0, 1, 2, \ldots\}, \qquad \mathcal{F} = \mathcal{P}(\Omega) = 2^{\Omega}.$$

For each of the following set functions, determine whether $(\Omega, \mathcal{F}, P)$ is a probability space. Justify your answer by checking the probability-measure axioms.

(i) For $\lambda > 0$, define

$$P(A) = e^{-\lambda} \sum_{x \in A} \frac{\lambda^x}{x!}, \qquad A \in \mathcal{F}.$$

(ii) For $0 < p < 1$, define

$$P(A) = \sum_{x \in A} p(1-p)^x, \qquad A \in \mathcal{F}.$$

(iii) Define

$$P(A) = \begin{cases} 0, & \text{if } A \text{ has finitely many elements,} \\ 1, & \text{if } A \text{ has infinitely many elements,} \end{cases} \qquad A \in \mathcal{F}.$$

Solution.

For a set function $P : \mathcal{F} \to [0, 1]$ to be a probability measure, it must satisfy:

1. $P(A) \geq 0$ for every $A \in \mathcal{F}$;
2. $P(\Omega) = 1$;
3. for every sequence of pairwise disjoint sets $A_1, A_2, \ldots \in \mathcal{F}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

A useful general observation.

Suppose that $q_x \geq 0$ for every $x \in \Omega$ and

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\sum_{x=0}^{\infty} q_x = 1.$$

Then

$$P(A) = \sum_{x \in A} q_x$$

defines a probability measure on $\mathcal{P}(\Omega)$.

Indeed, $P(A) \geq 0$, and

$$P(\Omega) = \sum_{x=0}^{\infty} q_x = 1.$$

If $A_1, A_2, \ldots$ are pairwise disjoint, then every element of their union occurs in exactly one of the sets. Therefore,

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= \sum_{x \in \bigcup_{n=1}^{\infty} A_n} q_x \\ &= \sum_{n=1}^{\infty} \sum_{x \in A_n} q_x \\ &= \sum_{n=1}^{\infty} P(A_n). \end{aligned}$$

We now apply this observation.

Part (i).

Define

$$q_x = e^{-\lambda} \frac{\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots.$$

Since $\lambda > 0$,

$$q_x \geq 0.$$

Furthermore, using the exponential-series identity,

$$e^{\lambda} = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!},$$

we obtain

$$\begin{aligned} P(\Omega) &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} e^{\lambda} \\ &= 1. \end{aligned}$$

Thus, the numbers

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$e^{-\lambda}\frac{\lambda^x}{x!}$$

form a probability mass function. Countable additivity follows from the general observation above. Hence,

$$\boxed{(\Omega, \mathcal{F}, P) \text{ is a probability space.}}$$

This is the probability measure associated with the Poisson distribution with parameter λ.

Part (ii).

Define

$$q_x = p(1-p)^x, \qquad x = 0, 1, 2, \ldots.$$

Since $0 < p < 1$,

$$q_x \geq 0.$$

Also, $0 < 1 - p < 1$, so the geometric-series formula gives

$$\sum_{x=0}^{\infty}(1-p)^x = \frac{1}{1-(1-p)} = \frac{1}{p}.$$

Therefore,

$$\begin{aligned} P(\Omega) &= \sum_{x=0}^{\infty} p(1-p)^x \\ &= p\sum_{x=0}^{\infty}(1-p)^x \\ &= p\left(\frac{1}{p}\right) \\ &= 1. \end{aligned}$$

Countable additivity again follows from the general observation. Consequently,

$$\boxed{(\Omega, \mathcal{F}, P) \text{ is a probability space.}}$$

This is a geometric probability distribution on $\{0, 1, 2, \ldots\}$.

Part (iii).

The set function is nonnegative, and because Ω is infinite,

$$P(\Omega) = 1.$$

However, it is not countably additive.

For every $n \geq 0$, let

$$A_n = \{n\}.$$

Each A_n is finite, so

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$P(A_n) = 0.$$

The singleton sets are pairwise disjoint and

$$\bigcup_{n=0}^{\infty} A_n = \Omega.$$

Therefore,

$$P\left(\bigcup_{n=0}^{\infty} A_n\right) = P(\Omega) = 1,$$

whereas

$$\sum_{n=0}^{\infty} P(A_n) = \sum_{n=0}^{\infty} 0 = 0.$$

Thus,

$$P\left(\bigcup_{n=0}^{\infty} A_n\right) \neq \sum_{n=0}^{\infty} P(A_n).$$

Hence, countable additivity fails, and

$$\boxed{(\Omega, \mathcal{F}, P) \text{ is not a probability space.}}$$

Therefore, the final conclusion is

$$\boxed{\text{Cases (i) and (ii) are probability spaces, but case (iii) is not.}}$$

□

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Probability Properties

Problem 4: Applications of probability properties

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $A, B, C \in \mathcal{F}$. Suppose that

$$\begin{aligned} P(A) = 0.6, \qquad P(B) = 0.5, \qquad P(C) = 0.4, \\ P(A \cap B) = 0.3, \quad P(A \cap C) = 0.2, \quad P(B \cap C) = 0.2, \end{aligned}$$

and

$$P(A \cap B \cap C) = 0.1.$$

Find the following probabilities:

(i)
$$P(A \cup B \cup C) \qquad \text{and} \qquad P(A^c \cap B^c \cap C^c).$$

(ii)
$$P\big((A \cup B) \cap C\big) \qquad \text{and} \qquad P\big(A \cup (B \cap C)\big).$$

(iii)
$$P\big((A^c \cup B^c) \cap C^c\big) \qquad \text{and} \qquad P\big((A^c \cap B^c) \cup C^c\big).$$

Solution.

Part (i).

Using the inclusion–exclusion formula for three events,

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C). \end{aligned}$$

Substituting the given values,

$$\begin{aligned} P(A \cup B \cup C) &= 0.6 + 0.5 + 0.4 \\ &\quad - 0.3 - 0.2 - 0.2 + 0.1 \\ &= 0.9. \end{aligned}$$

Therefore,

$$\boxed{P(A \cup B \cup C) = 0.9.}$$

By De Morgan's law,

$$A^c \cap B^c \cap C^c = (A \cup B \cup C)^c.$$

Hence,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned} P(A^c \cap B^c \cap C^c) &= 1 - P(A \cup B \cup C) \\ &= 1 - 0.9 \\ &= 0.1. \end{aligned}$$

Thus,

$$\boxed{P(A^c \cap B^c \cap C^c) = 0.1.}$$

Part (ii).

By the distributive law,

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C).$$

The two events $A \cap C$ and $B \cap C$ need not be disjoint. Their intersection is

$$(A \cap C) \cap (B \cap C) = A \cap B \cap C.$$

Therefore,

$$\begin{aligned} P\big((A \cup B) \cap C\big) &= P(A \cap C) + P(B \cap C) \\ &\quad - P(A \cap B \cap C) \\ &= 0.2 + 0.2 - 0.1 \\ &= 0.3. \end{aligned}$$

Hence,

$$\boxed{P\big((A \cup B) \cap C\big) = 0.3.}$$

Next, using the addition rule,

$$\begin{aligned} P\big(A \cup (B \cap C)\big) &= P(A) + P(B \cap C) \\ &\quad - P\big(A \cap (B \cap C)\big). \end{aligned}$$

Since

$$A \cap (B \cap C) = A \cap B \cap C,$$

we obtain

$$\begin{aligned} P\big(A \cup (B \cap C)\big) &= 0.6 + 0.2 - 0.1 \\ &= 0.7. \end{aligned}$$

Therefore,

$$\boxed{P\big(A \cup (B \cap C)\big) = 0.7.}$$

Part (iii).

First consider

$$(A^c \cup B^c) \cap C^c.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

By De Morgan's law,

$$A^c \cup B^c = (A \cap B)^c.$$

Consequently,

$$\begin{aligned}(A^c \cup B^c) \cap C^c &= (A \cap B)^c \cap C^c \\ &= \big((A \cap B) \cup C\big)^c.\end{aligned}$$

Now,

$$\begin{aligned}P\big((A \cap B) \cup C\big) &= P(A \cap B) + P(C) - P(A \cap B \cap C) \\ &= 0.3 + 0.4 - 0.1 \\ &= 0.6.\end{aligned}$$

Therefore,

$$\begin{aligned}P\big((A^c \cup B^c) \cap C^c\big) &= 1 - 0.6 \\ &= 0.4.\end{aligned}$$

Hence,

$$\boxed{P\big((A^c \cup B^c) \cap C^c\big) = 0.4.}$$

For the second probability, let

$$E = (A^c \cap B^c) \cup C^c.$$

Taking its complement and applying De Morgan's law gives

$$\begin{aligned}E^c &= (A^c \cap B^c)^c \cap (C^c)^c \\ &= (A \cup B) \cap C.\end{aligned}$$

From part (ii),

$$P\big((A \cup B) \cap C\big) = 0.3.$$

Therefore,

$$\begin{aligned}P(E) &= 1 - P(E^c) \\ &= 1 - 0.3 \\ &= 0.7.\end{aligned}$$

Thus,

$$\boxed{P\big((A^c \cap B^c) \cup C^c\big) = 0.7.}$$

□

Problem 5: Exactly one event and related probability identities

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $A, B \in \mathcal{F}$.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(i) Show that the probability that exactly one of the events A or B occurs is

$$P(A) + P(B) - 2P(A \cap B).$$

(ii) Show that

$$\begin{aligned} P(A \cap B) - P(A)P(B) &= P(A)P(B^c) - P(A \cap B^c) \\ &= P(A^c)P(B) - P(A^c \cap B) \\ &= P\big((A \cup B)^c\big) - P(A^c)P(B^c). \end{aligned}$$

Solution.

Part (i).

Exactly one of A or B occurs when either

$$A \cap B^c$$

occurs or

$$A^c \cap B$$

occurs. Thus, the required event is

$$E = (A \cap B^c) \cup (A^c \cap B).$$

These two events are disjoint because

$$(A \cap B^c) \cap (A^c \cap B) = \varnothing.$$

Therefore,

$$P(E) = P(A \cap B^c) + P(A^c \cap B).$$

Since

$$A = (A \cap B) \dot{\cup} (A \cap B^c),$$

we have

$$P(A \cap B^c) = P(A) - P(A \cap B).$$

Similarly,

$$P(A^c \cap B) = P(B) - P(A \cap B).$$

Consequently,

$$\begin{aligned} P(E) &= P(A) - P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - 2P(A \cap B). \end{aligned}$$

Hence,

This file is meant for personal use by harikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\boxed{P(\text{exactly one of } A \text{ or } B) = P(A) + P(B) - 2P(A \cap B).}$$

Part (ii).

Define

$$\Delta = P(A \cap B) - P(A)P(B).$$

We show that each expression in the question is equal to Δ.

First,

$$P(B^c) = 1 - P(B)$$

and

$$P(A \cap B^c) = P(A) - P(A \cap B).$$

Therefore,

$$\begin{aligned} P(A)P(B^c) - P(A \cap B^c) &= P(A)\big(1 - P(B)\big) \\ &\quad - \big(P(A) - P(A \cap B)\big) \\ &= P(A \cap B) - P(A)P(B) \\ &= \Delta. \end{aligned}$$

Next,

$$P(A^c) = 1 - P(A)$$

and

$$P(A^c \cap B) = P(B) - P(A \cap B).$$

Thus,

$$\begin{aligned} P(A^c)P(B) - P(A^c \cap B) &= \big(1 - P(A)\big)P(B) \\ &\quad - \big(P(B) - P(A \cap B)\big) \\ &= P(A \cap B) - P(A)P(B) \\ &= \Delta. \end{aligned}$$

Finally, by De Morgan's law,

$$(A \cup B)^c = A^c \cap B^c.$$

Using the addition rule,

$$\begin{aligned} P\big((A \cup B)^c\big) &= 1 - P(A \cup B) \\ &= 1 - P(A) - P(B) + P(A \cap B). \end{aligned}$$

Also,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned} P(A^c)P(B^c) &= \big(1 - P(A)\big)\big(1 - P(B)\big) \\ &= 1 - P(A) - P(B) + P(A)P(B). \end{aligned}$$

Subtracting the two expressions gives

$$\begin{aligned} &P\big((A \cup B)^c\big) - P(A^c)P(B^c) \\ &\quad = P(A \cap B) - P(A)P(B) \\ &\quad = \Delta. \end{aligned}$$

Therefore,

$$\boxed{\begin{aligned} P(A \cap B) - P(A)P(B) &= P(A)P(B^c) - P(A \cap B^c) \\ &= P(A^c)P(B) - P(A^c \cap B) \\ &= P\big((A \cup B)^c\big) - P(A^c)P(B^c). \end{aligned}}$$

□

This file is meant for personal use by harikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Applications of Probability

Problem 6: Distance from the nearest side of a square

A point (X, Y) is chosen uniformly at random from the unit square

$$S = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 1\}.$$

Thus, for any region $R \subseteq S$,

$$P\big((X, Y) \in R\big) = \frac{\text{Area}(R)}{\text{Area}(S)}.$$

Find the probability that the distance from (X, Y) to the nearest side of the square does not exceed $1/3$.

Solution.

Let D denote the distance from the point (X, Y) to the nearest side of the unit square.
The four distances from (X, Y) to the four sides are

$$X, \qquad 1 - X, \qquad Y, \qquad 1 - Y.$$

Therefore,

$$D = \min\{X, 1 - X, Y, 1 - Y\}.$$

We want to find

$$P\left(D \le \frac{1}{3}\right).$$

It is easier to calculate the probability of the complementary event

$$D > \frac{1}{3}.$$

Step 1: Describe the complementary event.
The condition $D > 1/3$ means that the point is more than $1/3$ unit away from every side. Hence,

$$X > \frac{1}{3}, \qquad 1 - X > \frac{1}{3},$$

and

$$Y > \frac{1}{3}, \qquad 1 - Y > \frac{1}{3}.$$

The conditions involving X give

$$\frac{1}{3} < X < \frac{2}{3},$$

and the conditions involving Y give

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\frac{1}{3} < Y < \frac{2}{3}.$$

Thus, the event $D > 1/3$ corresponds to the central square

$$\left(\frac{1}{3}, \frac{2}{3}\right) \times \left(\frac{1}{3}, \frac{2}{3}\right).$$

Step 2: Calculate its area.

The side length of the central square is

$$\frac{2}{3} - \frac{1}{3} = \frac{1}{3}.$$

Therefore, its area is

$$\left(\frac{1}{3}\right)^2 = \frac{1}{9}.$$

Since the area of the unit square is 1,

$$P\left(D > \frac{1}{3}\right) = \frac{1}{9}.$$

The boundary lines have area zero, so strict or non-strict inequalities do not change the probability.

Step 3: Use the complement rule.

Therefore,

$$\begin{aligned} P\left(D \leq \frac{1}{3}\right) &= 1 - P\left(D > \frac{1}{3}\right) \\ &= 1 - \frac{1}{9} \\ &= \frac{8}{9}. \end{aligned}$$

Hence,

$$\boxed{P\left(\text{distance to the nearest side} \leq \frac{1}{3}\right) = \frac{8}{9}.}$$

□

Problem 7: Real roots of a randomly formed quadratic equation

Three numbers a, b, c are chosen independently and with replacement from the set

$$\{1, 2, 3, 4, 5, 6\}.$$

Find the probability that the quadratic equation

$$ax^2 + bx + c = 0$$

has at least one real root.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution.

Because the numbers are chosen with replacement, repetitions are allowed. Each ordered triple

$$(a, b, c) \in \{1, 2, 3, 4, 5, 6\}^3$$

is equally likely.

Step 1: Count the total number of outcomes.

There are 6 choices for each of a, b, c. Therefore, the total number of ordered triples is

$$6^3 = 216.$$

Step 2: Use the discriminant condition.

The quadratic equation

$$ax^2 + bx + c = 0$$

has at least one real root if and only if its discriminant is nonnegative:

$$b^2 - 4ac \geq 0.$$

Equivalently,

$$ac \leq \frac{b^2}{4}.$$

We count the possible ordered pairs (a, c) for each fixed value of b.

Step 3: Count the favorable choices.

For a fixed a, count the values $c \in \{1, \ldots, 6\}$ satisfying

$$ac \leq \frac{b^2}{4}.$$

b	Condition on ac	Counts for $a = 1, \ldots, 6$	Total
1	$ac \leq \frac{1}{4}$	0	0
2	$ac \leq 1$	1	1
3	$ac \leq \frac{9}{4}$	$2 + 1$	3
4	$ac \leq 4$	$4 + 2 + 1 + 1$	8
5	$ac \leq \frac{25}{4}$	$6 + 3 + 2 + 1 + 1 + 1$	14
6	$ac \leq 9$	$6 + 4 + 3 + 2 + 1 + 1$	17

For example, when $b = 5$, the condition is

$$ac \leq \frac{25}{4}.$$

Since ac is an integer, this is equivalent to

$$ac \leq 6.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

The possible numbers of choices for c, corresponding to $a = 1, 2, \ldots, 6$, are

$$6,\ 3,\ 2,\ 1,\ 1,\ 1.$$

Thus, for $b = 5$, there are

$$6 + 3 + 2 + 1 + 1 + 1 = 14$$

favorable ordered pairs (a, c).

Adding the totals for all six values of b, the total number of favorable triples is

$$0 + 1 + 3 + 8 + 14 + 17 = 43.$$

Step 4: Calculate the probability.

Therefore,

$$P(\text{at least one real root}) = \frac{\text{number of favorable triples}}{\text{total number of triples}} = \frac{43}{216}.$$

Hence,

$$\boxed{P\big(ax^2 + bx + c = 0 \text{ has real root(s)}\big) = \frac{43}{216}.}$$

Numerically,

$$\frac{43}{216} \approx 0.1991.$$

Thus, the probability is approximately 19.91%.

□

Conditional Probability

Problem 8: Constructing a conditional probability space

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $B \in \mathcal{F}$ and

$$P(B) > 0.$$

We are told that the outcome of the experiment belongs to B. Therefore, B becomes the new sample space. Define

$$\mathcal{F}_B = \{A \cap B : A \in \mathcal{F}\}.$$

Complete the following steps.

(i) Show that

$$\mathcal{F}_B \subseteq \mathcal{F} \qquad \text{and} \qquad B \in \mathcal{F}_B.$$

(ii) Let $C \in \mathcal{F}_B$. Write $C = A \cap B$ for some $A \in \mathcal{F}$, and show that its complement relative to B satisfies

$$B \setminus C = A^c \cap B \in \mathcal{F}_B.$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

(iii) Let $C_n \in \mathcal{F}_B$ for every $n \geq 1$. Show that

$$\bigcup_{n=1}^{\infty} C_n \in \mathcal{F}_B.$$

Hence, conclude that $\mathcal{F}_B$ is a sigma-field of subsets of B.

(iv) Define

$$P_B(C) = \frac{P(C)}{P(B)}, \qquad C \in \mathcal{F}_B.$$

Verify that P_B is a probability measure on $(B, \mathcal{F}_B)$.

(v) For $A \in \mathcal{F}$, define

$$P(A \mid B) = P_B(A \cap B).$$

Show that

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

(vi) Show that, for fixed B, the set function

$$A \longmapsto P(A \mid B), \qquad A \in \mathcal{F},$$

is a probability measure on $(\Omega, \mathcal{F})$.

Solution.

The main idea is that once we know the outcome lies in B, we treat B as the new sample space and renormalize the probabilities inside B.

Part (i): Showing that $\mathcal{F}_B \subseteq \mathcal{F}$ and $B \in \mathcal{F}_B$.

Every element of $\mathcal{F}_B$ has the form

$$A \cap B$$

for some $A \in \mathcal{F}$. Since $A, B \in \mathcal{F}$ and a sigma-field is closed under intersections,

$$A \cap B \in \mathcal{F}.$$

Therefore,

$$\boxed{\mathcal{F}_B \subseteq \mathcal{F}.}$$

Also, $\Omega \in \mathcal{F}$. Taking $A = \Omega$, we obtain

$$\Omega \cap B = B.$$

Hence,

$$\boxed{B \in \mathcal{F}_B.}$$

Notice that B, rather than Ω, is the entire sample space for $\mathcal{F}_B$.

Part (ii): Closure under complements relative to B.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Let $C \in \mathcal{F}_B$. Then there exists $A \in \mathcal{F}$ such that

$$C = A \cap B.$$

The complement of C in the new sample space B is $B \setminus C$. Therefore,

$$\begin{aligned} B \setminus C &= B \setminus (A \cap B) \\ &= B \cap (A \cap B)^c \\ &= B \cap (A^c \cup B^c) \\ &= (A^c \cap B) \cup (B^c \cap B) \\ &= A^c \cap B. \end{aligned}$$

Since $A \in \mathcal{F}$, we have $A^c \in \mathcal{F}$. Thus,

$$A^c \cap B \in \mathcal{F}_B.$$

Consequently,

$$\boxed{C \in \mathcal{F}_B \quad \Longrightarrow \quad B \setminus C \in \mathcal{F}_B.}$$

In particular,

$$\varnothing = B \setminus B \in \mathcal{F}_B.$$

Part (iii): Closure under countable unions.

Let

$$C_n \in \mathcal{F}_B, \qquad n = 1, 2, \ldots.$$

For each n, there exists $A_n \in \mathcal{F}$ such that

$$C_n = A_n \cap B.$$

Therefore,

$$\begin{aligned} \bigcup_{n=1}^{\infty} C_n &= \bigcup_{n=1}^{\infty} (A_n \cap B) \\ &= \left(\bigcup_{n=1}^{\infty} A_n \right) \cap B. \end{aligned}$$

Since $\mathcal{F}$ is a sigma-field,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

It follows that

$$\left(\bigcup_{n=1}^{\infty} A_n \right) \cap B \in \mathcal{F}_B.$$

Thus,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\boxed{\bigcup_{n=1}^{\infty} C_n \in \mathcal{F}_B.}$$

We have shown that:

- $B \in \mathcal{F}_B$;
- $\mathcal{F}_B$ is closed under complements relative to B;
- $\mathcal{F}_B$ is closed under countable unions.

Therefore,

$$\boxed{\mathcal{F}_B \text{ is a sigma-field of subsets of } B.}$$

Part (iv): Showing that P_B is a probability measure.

Define

$$P_B(C) = \frac{P(C)}{P(B)}, \qquad C \in \mathcal{F}_B.$$

This is well defined because $P(B) > 0$ and $\mathcal{F}_B \subseteq \mathcal{F}$.

We verify the three probability axioms.

1. Nonnegativity.

For every $C \in \mathcal{F}_B$,

$$P(C) \geq 0.$$

Since $P(B) > 0$,

$$P_B(C) = \frac{P(C)}{P(B)} \geq 0.$$

2. Normalization.

The entire new sample space is B. Therefore,

$$P_B(B) = \frac{P(B)}{P(B)} = 1.$$

3. Countable additivity.

Let $C_1, C_2, \ldots \in \mathcal{F}_B$ be pairwise disjoint. Since $\mathcal{F}_B \subseteq \mathcal{F}$, countable additivity of P gives

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} P(C_n).$$

Hence,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned} P_B\left(\bigcup_{n=1}^{\infty} C_n\right) &= \frac{P\left(\bigcup_{n=1}^{\infty} C_n\right)}{P(B)} \\ &= \frac{\sum_{n=1}^{\infty} P(C_n)}{P(B)} \\ &= \sum_{n=1}^{\infty} \frac{P(C_n)}{P(B)} \\ &= \sum_{n=1}^{\infty} P_B(C_n). \end{aligned}$$

Therefore,

$$\boxed{(B, \mathcal{F}_B, P_B) \text{ is a probability space.}}$$

Part (v): Obtaining the conditional-probability formula.

Let $A \in \mathcal{F}$. Then

$$A \cap B \in \mathcal{F}_B.$$

By definition,

$$P(A \mid B) = P_B(A \cap B).$$

Using the definition of P_B,

$$\begin{aligned} P(A \mid B) &= P_B(A \cap B) \\ &= \frac{P(A \cap B)}{P(B)}. \end{aligned}$$

Thus,

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}.}$$

Part (vi): Conditional probability as a probability measure.

For fixed B, define

$$Q_B(A) = P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad A \in \mathcal{F}.$$

We verify the probability axioms.

First,

$$Q_B(A) \geq 0$$

because $P(A \cap B) \geq 0$ and $P(B) > 0$.

Next,

This file is meant for personal use by harikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned} Q_B(\Omega) &= P(\Omega \mid B) \\ &= \frac{P(\Omega \cap B)}{P(B)} \\ &= \frac{P(B)}{P(B)} \\ &= 1. \end{aligned}$$

Finally, let $A_1, A_2, \ldots \in \mathcal{F}$ be pairwise disjoint. Then the sets

$$A_1 \cap B,\ A_2 \cap B, \ldots$$

are also pairwise disjoint. Therefore,

$$\begin{aligned} Q_B\left(\bigcup_{n=1}^{\infty} A_n\right) &= \frac{P\left(\left(\bigcup_{n=1}^{\infty} A_n\right) \cap B\right)}{P(B)} \\ &= \frac{P\left(\bigcup_{n=1}^{\infty} (A_n \cap B)\right)}{P(B)} \\ &= \frac{\sum_{n=1}^{\infty} P(A_n \cap B)}{P(B)} \\ &= \sum_{n=1}^{\infty} \frac{P(A_n \cap B)}{P(B)} \\ &= \sum_{n=1}^{\infty} Q_B(A_n). \end{aligned}$$

Hence,

$$\boxed{A \longmapsto P(A \mid B) \text{ is a probability measure on } (\Omega, \mathcal{F}).}$$

Important distinction.

The two probability measures have different domains:

$$P_B : \mathcal{F}_B \longrightarrow [0, 1],$$

where the sample space is B, whereas

$$P(\,\cdot \mid B) : \mathcal{F} \longrightarrow [0, 1],$$

where the sample space is still written as Ω.

They are connected by

$$\boxed{P(A \mid B) = P_B(A \cap B).}$$

In simple words, conditioning on B means:

$$\boxed{\text{Keep only the outcomes inside } B, \text{ then divide by } P(B).}$$

□

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Problem 9: Conditional Probability

Six cards are dealt at random, without replacement, from a standard deck of 52 cards. The deck contains 13 hearts and 39 non-heart cards.

Define the events

$$A = \{\text{all six cards in the hand are hearts}\}$$

and

$$B = \{\text{at least five cards in the hand are hearts}\}.$$

Find the conditional probability

$$P(A \mid B).$$

Solution.

We use the conditional-probability formula

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Since every hand in A contains six hearts, it certainly contains at least five hearts. Therefore,

$$A \subseteq B,$$

and hence

$$A \cap B = A.$$

Thus,

$$P(A \mid B) = \frac{P(A)}{P(B)}.$$

Step 1: Count all six-card hands.

The total number of unordered six-card hands is

$$\binom{52}{6}.$$

All these hands are equally likely.

Step 2: Find $P(A)$.

For event A, all six cards must be chosen from the 13 hearts. The number of such hands is

$$\binom{13}{6}.$$

Therefore,

This file is meant for personal use by harikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$P(A) = \frac{\binom{13}{6}}{\binom{52}{6}}.$$

Step 3: Find $P(B)$.

The event B occurs when the hand contains at least five hearts. Since the hand contains six cards, there are two mutually exclusive possibilities:

$$B = \{\text{exactly five hearts}\} \dot{\cup} \{\text{exactly six hearts}\}.$$

For exactly five hearts, we choose

$$\binom{13}{5}$$

hearts from the 13 available hearts and

$$\binom{39}{1}$$

non-heart card from the 39 non-heart cards. Therefore, the number of hands containing exactly five hearts is

$$\binom{13}{5}\binom{39}{1}.$$

The number of hands containing exactly six hearts is

$$\binom{13}{6}.$$

Consequently,

$$P(B) = \frac{\binom{13}{5}\binom{39}{1} + \binom{13}{6}}{\binom{52}{6}}.$$

Step 4: Calculate the conditional probability.

Substituting $P(A)$ and $P(B)$, we obtain

$$P(A \mid B) = \frac{\frac{\binom{13}{6}}{\binom{52}{6}}}{\frac{\binom{13}{5}\binom{39}{1} + \binom{13}{6}}{\binom{52}{6}}}$$

$$= \frac{\binom{13}{6}}{\binom{13}{5}\binom{39}{1} + \binom{13}{6}}.$$

Now,

$$\binom{13}{6} = 1716, \qquad \binom{13}{5} = 1287, \qquad \binom{39}{1} = 39.$$

Hence,

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$\begin{aligned} P(A \mid B) &= \frac{1716}{(1287)(39) + 1716} \\ &= \frac{1716}{51909} \\ &= \frac{4}{121}. \end{aligned}$$

Therefore,

$$\boxed{P(A \mid B) = \frac{4}{121} \approx 0.0331.}$$

Thus, given that the hand contains at least five hearts, the probability that all six cards are hearts is approximately

$$\boxed{3.31\%}.$$

□

Problem 10: Successive draws without replacement

An urn contains 4 red balls and 6 black balls. Two balls are drawn successively and at random, without replacement.

Find the probability that the first ball drawn is red and the second ball drawn is black.

Solution.

Define the events

$$A = \{\text{the first ball drawn is red}\}$$

and

$$B = \{\text{the second ball drawn is black}\}.$$

The required event is

$$A \cap B.$$

By the multiplication rule,

$$P(A \cap B) = P(A)P(B \mid A).$$

Step 1: Probability that the first ball is red.

Initially, the urn contains 10 balls, of which 4 are red. Therefore,

$$P(A) = \frac{4}{10}.$$

Step 2: Probability that the second ball is black, given that the first ball was red.

After a red ball is drawn, there are

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$9$$

balls remaining. Since the removed ball was red, all 6 black balls are still in the urn. Hence,

$$P(B \mid A) = \frac{6}{9}.$$

Step 3: Apply the multiplication rule.

Therefore,

$$\begin{aligned} P(A \cap B) &= P(A)P(B \mid A) \\ &= \frac{4}{10} \times \frac{6}{9} \\ &= \frac{24}{90} \\ &= \frac{4}{15}. \end{aligned}$$

Thus,

$$\boxed{P(\text{first red and second black}) = \frac{4}{15}.}$$

Alternative counting check.

The total number of possible ordered pairs of balls is

$$10 \times 9.$$

The number of favorable ordered pairs is

$$4 \times 6,$$

because the first ball can be any one of the 4 red balls and the second ball can be any one of the 6 black balls. Therefore,

$$\frac{4 \times 6}{10 \times 9} = \frac{4}{15},$$

which agrees with the multiplication-rule calculation.

□

Problem 11: Selecting an urn using a die

Urn U_1 contains 4 white balls and 6 black balls, while urn U_2 contains 6 white balls and 4 black balls. A fair six-sided die is rolled.

- Urn U_1 is selected if the die shows 5 or 6.
- Urn U_2 is selected if the die shows $1, 2, 3$, or 4.

One ball is then drawn at random from the selected urn. Find the probability that the ball drawn is white.

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Solution.

Define the events

$$W = \{\text{the ball drawn is white}\},$$

$$E_1 = \{\text{urn } U_1 \text{ is selected}\},$$

and

$$E_2 = \{\text{urn } U_2 \text{ is selected}\}.$$

Exactly one of the two urns is selected. Therefore, E_1 and E_2 are mutually exclusive and exhaustive:

$$E_1 \cap E_2 = \varnothing, \qquad E_1 \cup E_2 = \Omega.$$

Thus, $\{E_1, E_2\}$ forms a partition of the sample space.

Step 1: Find the urn-selection probabilities.

Urn U_1 is selected when the die shows either 5 or 6. Therefore,

$$P(E_1) = \frac{2}{6} = \frac{1}{3}.$$

Urn U_2 is selected when the die shows one of $1, 2, 3$, or 4. Hence,

$$P(E_2) = \frac{4}{6} = \frac{2}{3}.$$

As a check,

$$P(E_1) + P(E_2) = \frac{1}{3} + \frac{2}{3} = 1.$$

Step 2: Find the conditional probability of drawing a white ball from each urn.

Urn U_1 contains 4 white balls out of 10 balls. Therefore,

$$P(W \mid E_1) = \frac{4}{10} = \frac{2}{5}.$$

Urn U_2 contains 6 white balls out of 10 balls. Therefore,

$$P(W \mid E_2) = \frac{6}{10} = \frac{3}{5}.$$

Step 3: Apply the law of total probability.

The event W can occur in two mutually exclusive ways:

$$W = (W \cap E_1) \dot{\cup} (W \cap E_2).$$

Therefore,

$$P(W) = P(W \cap E_1) + P(W \cap E_2).$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Using the multiplication rule,

$$P(W \cap E_i) = P(E_i)P(W \mid E_i).$$

Hence, the law of total probability gives

$$\begin{aligned} P(W) &= P(E_1)P(W \mid E_1) + P(E_2)P(W \mid E_2) \\ &= \frac{1}{3} \cdot \frac{2}{5} + \frac{2}{3} \cdot \frac{3}{5} \\ &= \frac{2}{15} + \frac{6}{15} \\ &= \frac{8}{15}. \end{aligned}$$

Therefore,

$$\boxed{P(\text{the drawn ball is white}) = \frac{8}{15}.}$$

Numerically,

$$\frac{8}{15} \approx 0.5333.$$

This value lies between 0.4 and 0.6, the white-ball probabilities in the two urns. It is closer to 0.6 because urn U_2 is selected more often.

□

Independent Events

Problem 12: Pairwise independence, mutual independence, and complements

Part A: Pairwise versus mutual independence

Let

$$\Omega = \{1, 2, 3, 4\}, \qquad \mathcal{F} = \mathcal{P}(\Omega),$$

and suppose that all four outcomes are equally likely:

$$P(\{i\}) = \frac{1}{4}, \qquad i = 1, 2, 3, 4.$$

Consider the events

$$A = \{1, 4\}, \qquad B = \{2, 4\}, \qquad C = \{3, 4\}.$$

(i) Calculate $P(A)$, $P(B)$, $P(C)$ and all three pairwise-intersection probabilities. Show that A, B, C are pairwise independent.

(ii) Calculate $P(A \cap B \cap C)$, and determine whether A, B, C are mutually independent.

Part B: Independence and complements

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Let E and F be independent events in an arbitrary probability space. Prove that each of the following pairs is also independent:

$$E^c \text{ and } F, \qquad E \text{ and } F^c, \qquad E^c \text{ and } F^c.$$

Solution.

Part A: Pairwise versus mutual independence

Because all four outcomes are equally likely, for every $D \subseteq \Omega$,

$$P(D) = \frac{|D|}{4}.$$

Step 1: Calculate the individual probabilities.

Each of A, B, C contains two outcomes. Therefore,

$$P(A) = \frac{|A|}{4} = \frac{2}{4} = \frac{1}{2},$$

and similarly,

$$P(B) = P(C) = \frac{1}{2}.$$

Thus,

$$\boxed{P(A) = P(B) = P(C) = \frac{1}{2}.}$$

Step 2: Calculate the pairwise intersections.

We have

$$A \cap B = \{1,4\} \cap \{2,4\} = \{4\},$$

$$A \cap C = \{1,4\} \cap \{3,4\} = \{4\},$$

and

$$B \cap C = \{2,4\} \cap \{3,4\} = \{4\}.$$

Therefore,

$$P(A \cap B) = P(A \cap C) = P(B \cap C) = P(\{4\}) = \frac{1}{4}.$$

Now,

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap B).$$

Similarly,

$$P(A)P(C) = \frac{1}{4} = P(A \cap C),$$

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

and

$$P(B)P(C) = \frac{1}{4} = P(B \cap C).$$

Thus, each pair satisfies the independence condition. Hence,

$$\boxed{A, B, C \text{ are pairwise independent.}}$$

Step 3: Check mutual independence.

For three events to be mutually independent, we must have both pairwise independence and

$$P(A \cap B \cap C) = P(A)P(B)P(C).$$

In this example,

$$A \cap B \cap C = \{4\}.$$

Therefore,

$$P(A \cap B \cap C) = P(\{4\}) = \frac{1}{4}.$$

On the other hand,

$$P(A)P(B)P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since

$$\frac{1}{4} \neq \frac{1}{8},$$

we have

$$P(A \cap B \cap C) \neq P(A)P(B)P(C).$$

Consequently,

$$\boxed{A, B, C \text{ are not mutually independent.}}$$

This example proves that

$$\boxed{\text{Pairwise independence does not imply mutual independence.}}$$

Part B: Independence and complements

Suppose that E and F are independent. Then

$$P(E \cap F) = P(E)P(F).$$

We prove the three required results separately.

1. Independence of E^c and F.

The event F can be written as the disjoint union

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

$$F = (E \cap F) \dot{\cup} (E^c \cap F).$$

Therefore,

$$P(F) = P(E \cap F) + P(E^c \cap F).$$

It follows that

$$\begin{aligned} P(E^c \cap F) &= P(F) - P(E \cap F) \\ &= P(F) - P(E)P(F) \\ &= \big(1 - P(E)\big)P(F) \\ &= P(E^c)P(F). \end{aligned}$$

Hence,

$$\boxed{E^c \text{ and } F \text{ are independent.}}$$

2. Independence of E and F^c.

The event E can be written as the disjoint union

$$E = (E \cap F) \dot{\cup} (E \cap F^c).$$

Therefore,

$$P(E) = P(E \cap F) + P(E \cap F^c).$$

Thus,

$$\begin{aligned} P(E \cap F^c) &= P(E) - P(E \cap F) \\ &= P(E) - P(E)P(F) \\ &= P(E)\big(1 - P(F)\big) \\ &= P(E)P(F^c). \end{aligned}$$

Hence,

$$\boxed{E \text{ and } F^c \text{ are independent.}}$$

3. Independence of E^c and F^c.

By De Morgan's law,

$$E^c \cap F^c = (E \cup F)^c.$$

Therefore,

$$\begin{aligned} P(E^c \cap F^c) &= 1 - P(E \cup F) \\ &= 1 - \big(P(E) + P(F) - P(E \cap F)\big). \end{aligned}$$

Since E and F are independent,

$$P(E \cap F) = P(E)P(F).$$

This file is meant for personal use by harikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.

Hence,

$$\begin{aligned} P(E^c \cap F^c) &= 1 - P(E) - P(F) + P(E)P(F) \\ &= \big(1 - P(E)\big)\big(1 - P(F)\big) \\ &= P(E^c)P(F^c). \end{aligned}$$

Consequently,

$$\boxed{E^c \text{ and } F^c \text{ are independent.}}$$

Therefore, if E and F are independent, independence is preserved when either one or both events are replaced by their complements:

$$\boxed{E^c \text{ and } F, \qquad E \text{ and } F^c, \qquad E^c \text{ and } F^c \text{ are independent.}}$$

□

This file is meant for personal use by parikshithrnayak@gmail.com only.
Sharing or publishing the contents in part or full is liable for legal action.